DOMINATING AN FPPF COVER BY ONE SURJECTIVE INTEGRAL MORPHISM WITH LOCALLY FACTORIZING OPENS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 13733-13755, immediately following the statement at 13725-13731.

Editorial note. Two same-chapter pointers (dominate-fppf, extend-integral-surjective-morphisms) and two cross-chapter Limits pointer pairs are recorded in assembly.external_references.

Editorial note. Related card S9-morph-dominate-fppf is not present: the local normal form lemma itself was not counted by this lane; only this globalized version is.

Allowed standard prerequisites (editorial).

- Local normal form of an fppf cover. Lemma lemma-dominate-fppf of this chapter, cited at proof lines 13733-13735. Supplies the opens U_j of S, the morphisms W_j -> U_j, their opens W_{j,k} and the pieces T_{j,k} -> W_{j,k} that the construction extends.
- Extending a morphism to a surjective integral morphism. Lemma lemma-extend-integral-surjective-morphisms of this chapter, cited at proof lines 13736-13737 and 13738-13739. Extends W_j -> U_j to an integral surjective end over S and T_{j,k} -> W_{j,k} to one over the first extension.
- Infinite products of schemes exist. Limits, Lemma limits-lemma-infinite-product, cited at proof lines 13740-13742. Justifies forming the product of all T-bar_{j,k} over W-bar_j and of all T-bar_j over S.
- Integral surjectivity is preserved by infinite products. Limits, Lemmas limits-lemma-infinite-product-integral and limits-lemma-infinite-product-surjective, cited at proof lines 13753-13755. Concludes S' -> S is surjective integral from the product construction.

Source reference. Original lemma 05WN: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13649–13649. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0CNX: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13686–13686. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0CNI: focused-seeds/geometry/S9-morph-dominate-fppf-integral/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0CNK: focused-seeds/geometry/S9-morph-dominate-fppf-integral/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0CNJ: focused-seeds/geometry/S9-morph-dominate-fppf-integral/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $\{S_i \to S\}_{i \in I}$ be an fppf covering. Then there exists a surjective integral morphism $S' \to S$ and an open covering $S' = \bigcup U'_\alpha$ such that for each α the morphism $U'_\alpha \to S$ factors through $S_i \to S$ for some i.*

Proof. Choose $S = \bigcup U_j$, $W_j \to U_j$, $W_j = \bigcup W_{j,k}$, and $T_{j,k} \to W_{j,k}$ as in Lemma 05WN. By Lemma 0CNX we can extend $W_j \to U_j$ to a surjective integral morphism $\overline{W}_j \to S$. After this we can extend $T_{j,k} \to W_{j,k}$ to a surjective integral morphism $\overline{T}_{j,k} \to \overline{W}_j$. We set $\overline{T}_j$ equal to the product of all the schemes $\overline{T}_{j,k}$ over $\overline{W}_j$ (Limits, Lemma 0CNI). Then we set S' equal to the product of all the schemes $\overline{T}_j$ over S. If $x \in S'$, then there is a j such that the image of x in S lies in U_j. Hence there is a k such that the image of x under the projection $S' \to \overline{W}_j$ lies in $W_{j,k}$. Hence under the projection $S' \to \overline{T}_j \to \overline{T}_{j,k}$ the point x ends up in $T_{j,k}$. And $T_{j,k} \to S$ factors through S_i for some i. Finally, the morphism $S' \to S$ is integral and surjective by Limits, Lemmas 0CNK and 0CNJ. □